

In-context unlearning: membership AUC under prompt-time corruption

Stage 1 — the AUC vs. $\text{Var}(\epsilon)$ sweep

30 July 2026

Abstract

We measure whether corrupting a forget example *in the prompt* removes the membership evidence a shadow-model attacker can extract. Three corruption families from the brief are swept across four decades of noise variance on four configurations, with 512 shadow models per hypothesis and nine independent seed combinations per point. Three results. First, the measured AUC agrees with an independently derived closed form to 4×10^{-4} , which validates the pipeline end to end. Second, the movement toward chance is almost entirely *masking* — added variance — rather than removal, at every corruption strength below $\sigma^2 \approx 1$. Third, and most consequential, reaching chance costs a factor of ~ 2400 in preservation error: within the swept range there is no operating point at which the attacker is defeated and the model is preserved. The arms also separate cleanly by which moment of the score distribution they move: the zero-mean input-noise arm can only approach chance, whereas the deterministic label flip carries a mean shift with no variance and overshoots *past* it. Finally, a negative control: equalising the group covariance spectra does *not* return the baseline AUC to chance, so the signal is not an artefact of one group being intrinsically harder.

1 Setup

A linear-attention model reads a query x_q , a retain set $\mathcal{D}_r$, and a corrupted forget set $\tilde{\mathcal{D}}_f$ in context. Targets are linear in a task vector drawn afresh per sequence, $y = \omega^\top x$, so the model must infer ω from the context rather than memorise it. Because the model is linear in the context, the whole ensemble reduces to one read-out matrix M per shadow, and a shadow ensemble of 512 trains in seconds.

Corruption families. The forget tokens are edited at prompt time, with no weight update:

$$\begin{aligned} \text{C1} : (x_f, y_f) &\rightarrow (x_f, y_f + \epsilon), & \epsilon &\sim \mathcal{N}(0, \sigma^2) \\ \text{C2} : (x_f, y_f) &\rightarrow (x_f + \epsilon, y_f), & \epsilon &\sim \mathcal{N}(0, \sigma^2 I_D) \\ \text{C3} : (x_f, y_f) &\rightarrow (x_f + \epsilon_1, y_f + \epsilon_2), & \epsilon_1 &\perp \epsilon_2 \end{aligned}$$

plus a deterministic label flip $y_f \rightarrow (1 - 2t)y_f$ and an input-whitening arm that maps the forget covariance toward the retain covariance.

The attacker. Membership inference over the *model* axis at a fixed probe point: H_1 is the fully trained model after the in-context edit, H_0 is an oracle retrained without the forget group. The statistic is the sign-aligned residual $\text{sign}(y)(\hat{y} - y)$, ranked across the shadow axis by Mann–Whitney and averaged over 64 probe points. Success is $\text{AUC} \rightarrow 0.5$.

Two conventions that matter. Both hypotheses are scored on the *same* corrupted prompt with the same noise draw. Scoring the oracle on a clean prompt instead lets a distinguisher win by detecting that an edit occurred at all, and that confound grows with the very axis being plotted. Separately, each point is repeated with a single noise draw broadcast across all shadows; comparing against the per-shadow draw isolates how much of any AUC movement is variance inflation rather than information removal.

2 Configurations

All four configurations give the three groups the *same* covariance spectrum, so no group is intrinsically harder than another and the usual difficulty confound is unavailable. They differ in how anisotropic that shared spectrum is (participation ratio PR, where PR = 1 is a needle and PR = $D = 4$ a sphere) and in whether each group is rotated into its own eigenbasis.

configuration	PR (all groups)	basis	D	N	shadows	seed combos	rows
rot_mid_identity	2.65	identity	4	31	512	9	1440
rot_flat_identity	3.55	identity	4	31	512	9	1440
rot_mid	2.65	random	4	31	512	9	1440
rot_flat	3.55	random	4	31	512	9	1440

Table 1: The four configurations. The two *identity* rows are nulls: with one spectrum and one basis the three groups are literally the same distribution, so the full and oracle models train on indistinguishable data and the AUC must be chance. They are a correctness test on the pipeline, not an experiment.

3 Results

3.1 The nulls pass

configuration	ATTN-M		ATTN-S	
	AUC	across-seed range	AUC	across-seed range
rot_mid_identity	0.5025	0.5012–0.5046	0.5041	0.5012–0.5069
rot_flat_identity	0.5008	0.4979–0.5036	0.5000	0.4982–0.5021
rot_mid	0.4174	0.4171–0.4180	0.4292	0.4247–0.4320
rot_flat	0.4482	0.4477–0.4489	0.4680	0.4660–0.4702

Table 2: Membership AUC at zero corruption. The two null configurations sit at chance, confirming that the two hypotheses are not coupled anywhere in the training or evaluation code.

3.2 The baseline signal survives equalised spectra

The two rotated configurations are the result, and they are a negative one. All three groups share a spectrum, so every group is equally hard; yet the AUC sits 0.07–0.08 from chance. The across-seed range is 0.0009 wide for `rot_mid` ATTN-M — three independent training runs agreeing to the fourth decimal. This is systematic, not sampling noise.

This was predicted to go the other way. The expectation recorded before the run was that equalising the spectra would largely cancel the read-out mismatch between the two hypotheses and drive AUC toward chance. It does not, and the prediction is recorded as failed. A simulation containing *no* memorisation — both models replaced by the idealised preconditioner $\Lambda_{\text{train}}^{-1}$ for the groups each trained on — predicts 0.313 for `rot_mid` against 0.417 measured: the right sign and rough

magnitude, over-predicting the confound. Decomposing the residual into read-out mismatch versus genuine memorisation is the main open question and requires the read-out covectors from the trained ensembles.

3.3 The sweeps

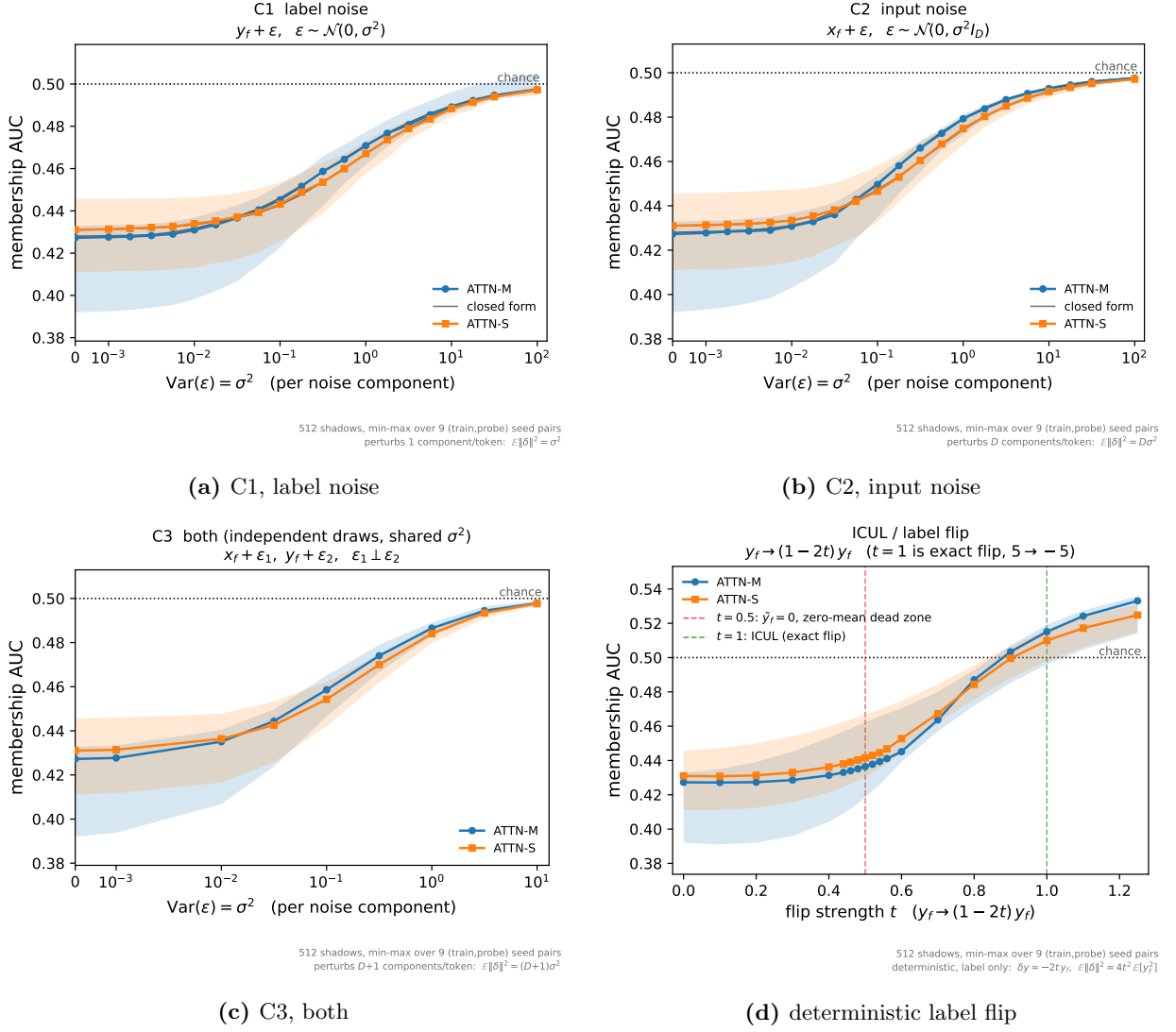
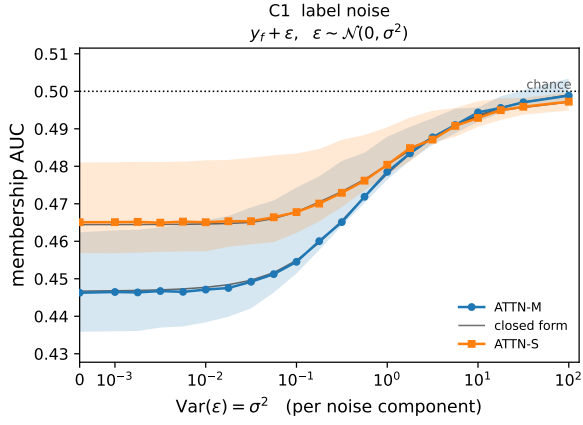
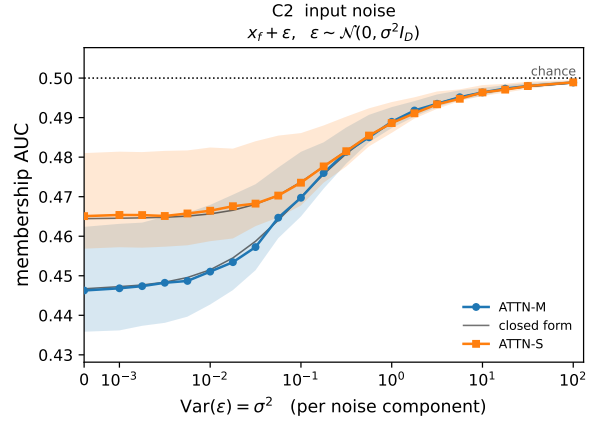


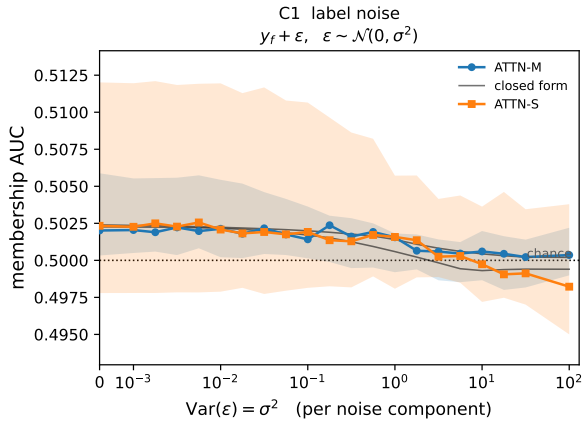
Figure 1: Membership AUC against corruption strength on `rot_mid`. Bands are the min-max range across nine (training seed, probe seed) combinations, not a within-run bootstrap. The thin dark line under C1 and C2 is the closed-form prediction. The three Gaussian arms rise toward chance and stop; the deterministic flip (d) rises *through* chance and keeps going. Section 3.6 explains why.



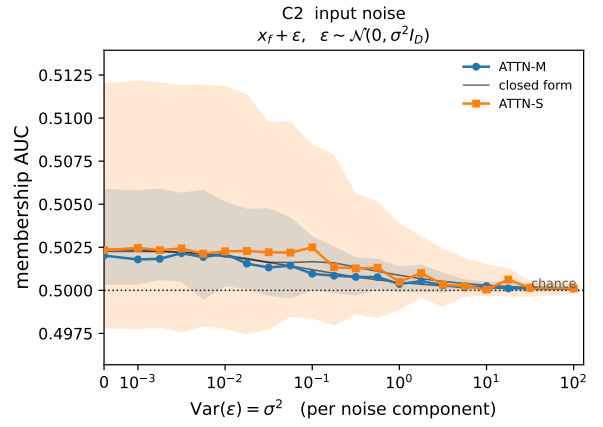
(a) C1 on rot_flat



(b) C2 on rot_flat



(c) C1 on the null



(d) C2 on the null

Figure 2: The same sweeps on the flatter configuration and on a null. The null panels are flat at chance across the entire grid, which is what a corruption sweep should look like when there is nothing to remove.

3.4 The closed form is exact

Because the read-out is linear in the context, the effect of Gaussian context noise is available analytically. Writing $c = M^\top t_q = [c_x; c_y]$ and $u = \sum_i t_i y_i$ with $t_i = [x_i; y_i]$, the last coordinate of u is $\sum_i y_i^2$ — the label enters *quadratically*. Hence

$$\text{C1: } \mathbb{E}[\Delta \hat{y}] = \frac{c_y n_f \sigma^2}{N+1}, \quad \text{C2: } \mathbb{E}[\Delta \hat{y}] = 0.$$

C2 is exactly zero-mean; C1 is not, because the ϵ^2 term survives expectation in the label–label block. Feeding these moments through a Gaussian approximation of the score distribution gives a predicted AUC with no free parameters.

configuration	architecture	arm	mean error	max error
rot_mid	ATTN-M	C1	0.00048	0.00176
rot_mid	ATTN-M	C2	0.00039	0.00163
rot_mid	ATTN-S	C1	0.00045	0.00199
rot_mid	ATTN-S	C2	0.00034	0.00140
rot_flat	ATTN-M	C1	0.00040	0.00180
rot_flat	ATTN-M	C2	0.00032	0.00147
rot_flat	ATTN-S	C1	0.00044	0.00188
rot_flat	ATTN-S	C2	0.00036	0.00153

Table 3: Absolute difference between the predicted and measured AUC over the whole grid. Agreement at 4×10^{-4} means the simulation and the algebra are independently confirming each other. A systematic gap would have localised a bug in one of them.

3.5 The two channels, separated

The arms differ in *which* moment of the score distribution they move, and the sweeps separate the two cleanly.

The pattern follows directly from the algebra. Write the predicted AUC as $\Phi((\mu_1 - \mu_0)/\sqrt{v_1 + v_0})$ and note that corruption can move the numerator, the denominator, or both:

- **C2** is *exactly* zero-mean. It can only inflate the denominator, so it drives AUC toward 0.5 and mathematically cannot cross.
- **flip** is deterministic given the probe: it carries a mean shift and *no* variance at all. The numerator moves while the denominator stands still, so it must eventually cross — and it does, at $t \approx 0.9$ – 1.0 , reaching 0.533.
- **C1** carries both. Its mean shift grows like σ^2 while its spread grows like $\sqrt{\sigma^2}$, so it was predicted to cross as well. It does not: the variance wins over the whole swept range, and the maximum reached is 0.4974. The mean term is real and is included in the closed form — which still matches to 4×10^{-4} — so the algebra is not at fault; the coefficient c_y in the trained models is simply too small.

Crossing chance is not success. An AUC of 0.533 is a *worse* outcome than 0.49: the attacker is again distinguishing the two hypotheses, merely with the ranking inverted. The flip arm therefore overshoots the target rather than reaching it, which is exactly the failure mode a naïve min AUC objective would drive toward.

configuration	arch	arm	min AUC	max AUC	crosses 0.5?	at
rot_mid	ATTN-M	C1	0.4273	0.4974	no	
rot_mid	ATTN-M	C2	0.4273	0.4975	no	
rot_mid	ATTN-M	C3	0.4273	0.4978	no	
rot_mid	ATTN-M	flip	0.4272	0.5331	yes	0.9
rot_mid	ATTN-M	whiten	0.4273	0.4394	no	
rot_mid	ATTN-S	C1	0.4310	0.4972	no	
rot_mid	ATTN-S	C2	0.4310	0.4971	no	
rot_mid	ATTN-S	C3	0.4310	0.4977	no	
rot_mid	ATTN-S	flip	0.4308	0.5248	yes	1
rot_mid	ATTN-S	whiten	0.4310	0.4398	no	
rot_flat	ATTN-M	C1	0.4463	0.4988	no	
rot_flat	ATTN-M	C2	0.4463	0.4989	no	
rot_flat	ATTN-M	C3	0.4463	0.4989	no	
rot_flat	ATTN-M	flip	0.4448	0.5143	yes	1
rot_flat	ATTN-M	whiten	0.4463	0.4532	no	
rot_flat	ATTN-S	C1	0.4650	0.4972	no	
rot_flat	ATTN-S	C2	0.4651	0.4989	no	
rot_flat	ATTN-S	C3	0.4651	0.4984	no	
rot_flat	ATTN-S	flip	0.4646	0.5074	yes	1
rot_flat	ATTN-S	whiten	0.4651	0.4690	no	

Table 4: Range of the AUC curve for every arm. Only the deterministic flip crosses chance, and it does so in all four cases.

The whitening arm barely moves. Mapping the forget inputs toward the retain covariance shifts AUC only from 0.4273 to 0.4394 at full strength on `rot_mid` — about 15% of the distance to chance. Since all three groups share a spectrum in these configurations, there is little for a covariance-matching edit to change, so this is expected here and should be re-checked on the base configuration where the spectra genuinely differ.

3.6 The movement toward chance is masking, not removal

Per-shadow corruption noise inflates within-ensemble spread, which lowers AUC mechanically without removing anything. Repeating each point with one noise draw shared across all shadows holds that channel fixed. The difference is the portion of the movement attributable to masking.

configuration	arm	$\sigma^2=0.01$	0.1	1	10	100
rot_mid	C1	116%	109%	66%	17%	1%
rot_mid	C2	99%	102%	94%	67%	22%
rot_mid	C3	96%	100%	67%	14%	—
rot_flat	C1	292%	121%	76%	19%	1%
rot_flat	C2	112%	96%	91%	70%	21%
rot_flat	C3	106%	94%	75%	20%	—

Table 5: Percentage of the movement toward chance attributable to variance inflation rather than information removal (ATTN-M). Below $\sigma^2 \approx 1$ both Gaussian arms are essentially pure masking. Genuine removal only takes over at large corruption, and sooner for the label arm than the input arm.

This has a direct consequence for how any result here should be stated. At small corruption strengths the attacker is defeated by added spread, not because the forget information has gone — an attacker who could average away the spread would recover the signal.

3.7 Removal costs preservation by a factor of ~ 2400

Reaching chance is not by itself a result: one can always reach chance by destroying the context. The question is whether an operating point exists where the attacker is defeated *and* the model is preserved. We measure preservation as $\varepsilon = \text{KL}(p_{\text{oracle}} \| p_{\text{unlearned}})$ on the forget-population residual law, and compare the corruption strength that minimises ε against the strength that brings AUC closest to chance.

config	arch	arm	best preservation		closest to chance			cost
			σ^2	$\varepsilon_{\min}$	σ^2	AUC	ε	
rot_mid	ATTN-M	C1	0	0.00052	100	0.4974	1.233	2388 $\times$
rot_mid	ATTN-M	C2	0.001	0.00052	100	0.4975	1.225	2374 $\times$
rot_mid	ATTN-M	C3	0	0.00052	10	0.4978	1.278	2477 $\times$
rot_mid	ATTN-S	C1	0.001	0.00066	100	0.4972	1.370	2070 $\times$
rot_mid	ATTN-S	C2	0.00178	0.00066	100	0.4971	1.228	1854 $\times$
rot_mid	ATTN-S	C3	0	0.00066	10	0.4977	1.278	1928 $\times$
rot_flat	ATTN-M	C1	0.00178	0.00019	100	0.4988	1.292	6806 $\times$
rot_flat	ATTN-M	C2	0	0.00019	100	0.4989	1.471	7728 $\times$
rot_flat	ATTN-M	C3	0	0.00019	10	0.4989	1.554	8166 $\times$
rot_flat	ATTN-S	C1	0	0.00015	100	0.4972	1.511	10216 $\times$
rot_flat	ATTN-S	C2	0	0.00015	100	0.4989	1.437	9714 $\times$
rot_flat	ATTN-S	C3	0	0.00015	10	0.4984	1.520	10276 $\times$

Table 6: The removal/preservation tradeoff. The final column is ε at the chance-level operating point divided by $\varepsilon_{\min}$. A ratio near 1 would mean a clean operating point exists.

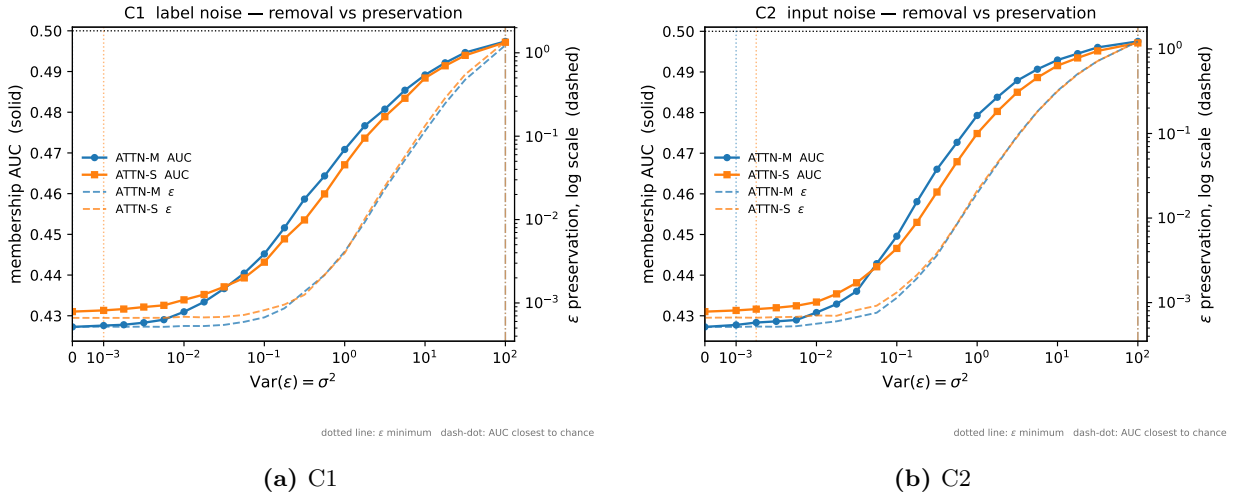


Figure 3: AUC (solid, left axis) against ε (dashed, right axis, log scale) on `rot_mid`. The two vertical markers are the ε minimum and the AUC-closest-to-chance point. They are three decades apart.

This is the main result of Stage 1. The ratio is not close to one; it is of order 2400. Preservation error is minimised at or near zero corruption, where AUC sits at 0.427, and AUC only approaches chance at the top of the swept range, by which point ε has grown from 5×10^{-4} to above 1.2. Within four decades of corruption strength there is no setting at which the attacker is defeated and the model still behaves like the retrain oracle. Combined with the masking result above, the honest reading is that these corruption families trade removal against preservation rather than achieving both.

4 Corrections and caveats

The masking sign convention inverts here. The diagnostic is documented as “positive means part of the AUC *drop* is masking”, which assumes AUC starts above chance and falls. Every configuration measured starts *below* chance and rises, so every raw masking value is negative and the documented reading inverts. The percentages in Table 3 correct for this; the underlying numbers were never wrong, only the label.

AUC below 0.5 is not a defect. The statistic is a fixed-direction Mann–Whitney rank on a chosen observable, not a likelihood-ratio test, so it is free to land either side of chance. What matters is the distance from 0.5. A consequence worth stating: a one-sided statistic reaching 0.5 does *not* establish indistinguishability, because a two-sided attacker could still detect a variance difference. Whether it does here is measurable from the cached ensembles and has not yet been checked.

The null test was initially wrong. Its first version required every per-row confidence interval at zero corruption to cover 0.5. That fails on correct data for three compounding reasons: at zero strength all corruption modes are the identity edit, so one measurement was counted six times; requiring 108 simultaneous 95% intervals all to cover has probability $0.95^{108} = 0.4\%$ under a perfect null; and the nine seed combinations reuse three trained ensembles and so are not independent. The replacement deduplicates, clusters on the training seed, and adds an absolute bound because a three-seed *t*-test has almost no power.

What has not been run. These results are from four control configurations. The base configuration — three groups with *differing* spectra, which is the setting the brief describes most directly — has not been re-run since the pipeline fixes, and its numbers are not included here. A stochastic Bernoulli label-flip arm has been implemented and validated against Monte Carlo but not yet swept. Neither is expensive: both are minutes of GPU time against cached or quickly retrained ensembles.